

CS & IT ENGINEERING

Discrete Mathematics

Combinatorics

DPP- 01



By- Mili Dhara Ma'am

[MCQ]

String of length 2 that contains a substring '00'



#Q. Let B_n represent the number of bit strings of length n that contain a substring "00"

Write the recurrence relation for B_n .

$$B_0 = 0, B_1 = 0, B_2 = 1$$

$$S = \underbrace{0 \ 0 \ 0/1 \ 0/1 \ 0/1 \ \dots \ 1}_{n-2}$$

$$B_n = B_{n-1} + B_{n-2} + 2^{n-2}$$

$\downarrow \qquad \qquad \downarrow \qquad \qquad \downarrow$
 $1 \qquad \qquad 01 \qquad \qquad 00$

☒ **A** $B_n = B_{n-1} + B_{n-2} + 2^{n-2}$

☐ **B** $B_n = B_{n-1} + B_{n-2} - 2^{n-2}$

☐ **C** $B_n = B_{n-1} - B_{n-2} + 2^{n-2}$

☐ **D** $B_n = B_{n-1} - B_{n-2} - 2^{n-2}$

#Q. A population of bacteria has initial count 200. After one hour, the population reached 220.

The population grows in such a way that the number of additional bacteria per hour, doubles each hour.

Write a recurrence relation ' b_n ' to describe the number of bacteria after n hours.

We know $b_0 = 200$ and $b_1 = 220$

A

$$b_n = 2b_{n-1} + 3b_{n-2}$$

B

$$b_n = 2b_{n-1} - 3b_{n-2}$$

C

$$b_n = 3b_{n-1} + 2b_{n-2}$$

D

$$b_n = 3b_{n-1} - 2b_{n-2}$$

$$b_0 = 200, \quad b_1 = 220$$

1 hr 1 hr

$$\underbrace{220 - 200}_{= 20}$$

$$b_2 - b_1 = 40$$

$$b_3 - b_2 = 80$$

$$b_n - b_{n-1} = 2(b_{n-1} - b_{n-2})$$

$$\Rightarrow b_n = 3b_{n-1} - 2b_{n-2}$$

#Q. Find the solution of recurrence relation using repeated substitution method.

$$a_n = 2a_{n-1} + 3 \quad n \geq 1$$

with initial terms $a_0 = 2$

$$a_0 = 2$$

$$\Rightarrow n - k = 0$$

$$\Rightarrow n = k$$

$$a_n = 2a_{n-1} + 3$$

$$= 2(2a_{n-2} + 3) + 3$$

$$\stackrel{3^{rd}}{\Rightarrow} = 2[2(2a_{n-3} + 3) + 3] + 3$$

$\vdots$ k times

$$= 2^k a_{n-k} + 2^{k-1} \times 3 + 2^{k-2} \times 3 + 2^{k-3} \times 3 \dots 2^0 \times 3$$

$$= 2^n a_0 + 3(2^{n-1} + 2^{n-2} + 2^{n-3} + \dots + 2^1 + 2^0)$$

$$= 2^{n+1} + 3(2^n - 1) = 2^{n+1} + 3 \cdot 2^n - 3 = 2^n(2 + 3) - 3$$

$$2^2 \times 3 + 2^1 \times 3 + 2^0 \times 3$$

$$\frac{a(r^n - 1)}{r - 1} = \frac{1(2^n - 1)}{2 - 1}$$

$$r = 2$$

A $5 \cdot 2^n + 3$

B $5 \cdot 2^n - 3$

C $3 \cdot 2^n - 5$

D $3 \cdot 2^n + 5$

[MCQ]

Put $n = n+3$

$$a_{n+3} = a_{n+2} + 8a_{n+1} - 12a_n$$

$$a_n = \left(\frac{1}{25} + \frac{2n}{5} \right) 2^n - \frac{1}{25} (-3)^n$$



#Q. Solve the recurrence relation $a_n = a_{n-1} + 8a_{n-2} - 12a_{n-3}$

subjected to the initial conditions $a_0 = 0, a_1 = 1, a_2 = 3$.

$$\phi(E) = E^3 - E^2 - 8E + 12$$

$$\Rightarrow \phi(t) = t^3 - t^2 - 8t + 12 = 0$$

$$(t-2)^2 (t+3) = 0$$

$$t = 2, 2, -3$$

A $a_n = \left(\frac{1}{25} + \frac{2n}{5} \right) 2^n - \left(\frac{2}{25} \right) (-3)^n$

$$a_n = (c_1 + c_2 n) 2^n + c_3 (-3)^n$$

$$a_0 = c_1 + c_3$$

$$\Rightarrow 0 = c_1 + c_3$$

$$c_1 = \frac{1}{25}, c_2 = \frac{2}{5},$$

$$c_3 = -\frac{1}{25}$$

B $a_n = \left(\frac{1}{25} + \frac{2n}{5} \right) 2^n - \left(\frac{1}{25} \right) (3)^n$

C $a_n = \left(\frac{1}{25} - \frac{2n}{5} \right) 2^n - \left(\frac{1}{25} \right) (-3)^n$

D $a_n = \left(\frac{1}{25} + \frac{2n}{5} \right) 2^n - \left(\frac{1}{25} \right) (-3)^n$

#Q. Solve the recurrence relation $a_n = 11 a_{n-1} - 30 a_{n-2}$
subjected to the initial conditions $a_0 = 1, a_1 = 5$

$$t^2 - 11t + 30 = 0$$

$$\Rightarrow (t-6)(t-5) = 0$$

$$t = 6, 5$$

$$a_n = c_1 (5)^n + c_2 (6)^n$$

$$a_0 = c_1 + c_2$$

$$\Rightarrow 1 = c_1 + c_2 \quad \text{--- (i)}$$

$$a_1 = 5c_1 + 6c_2$$

$$\Rightarrow 5 = 5c_1 + 6c_2 \quad \text{--- (ii)}$$

$$c_1 = 1, c_2 = 0$$

$$a_n = 5^n$$

☒ **A** The characteristic roots are 0 and 1.

☒ **B** $a_n = 5^n$

☒ **C** The characteristic roots are 5 and 6.

☐ **D** $a_n = 1 - 5^n$

[MCQ]

$$t^2 + 7t - 18 = 0$$

$$t = 2, -9 \checkmark (t-2)(t+9)$$

$$PS = \text{Sol of } \left(\frac{2^{n+2} + 1}{(t-2)(t+9)} \right) \text{PW}$$

#Q. Solve the recurrence relation $a_n + 7a_{n-1} - 18a_{n-2} = 2^n + 1$
subjected to the initial conditions $a_0 = 1, a_1 = 2$

$$a_n^{(H)} = C_1 (2)^n + C_2 (-9)^n$$

$$f(n) = 2^n + 1$$

$$F(n) = 2^{n+2} + 1 = 4 \cdot 2^n + 1$$

$$= \frac{4 \cdot 2^n}{(t-2)(t+9)} + \frac{1}{(t-2)(t+9)}$$

$$= 4 \times \frac{1}{2+9} \times \binom{n}{1} 2^{n-1} + \frac{1}{(-1)(10)}$$

$$= a_n = \frac{4n}{11} 2^{n-1} - \frac{1}{10}$$

☒ **A** $a_n = \frac{4n}{11} 2^{n-1} - \frac{1}{10}$

☐ **B** $a_n = \frac{4n}{11} 2^{n-1} + \frac{1}{10}$

☐ **C** $a_n = \frac{4n}{11} 2^{n-2} - \frac{1}{10}$

☐ **D** $a_n = \frac{3n}{11} 2^{n-1} - \frac{1}{10}$

[MCQ]

$$a_n = \left(1 - \frac{1}{3}n\right)3^n + {}^nC_2 3^{n-2}$$

#Q. Solve the recurrence relation $a_n = 6a_{n-1} - 9a_{n-2} + 3^n$ subjected to the initial conditions $a_0 = 1, a_1 = 2$

$$t^2 - 6t + 9 = 0$$

$$a_n^{(H)} = (C_1 + C_2 n)3^n$$

$$(t-3)^2$$

$$t = 3, 3$$

$$F(n) = 3^{n+2} = 9 \cdot 3^n$$

$$\frac{b^n}{\phi(E)}$$

$$a_n^{(P)} = 9 \cdot \text{sol of } \left(\frac{3^n}{\phi(E)} \right) [b=3]$$

$$\phi(b) = \phi(3) = 0$$

$$\phi(E) = (E-b)^m \cdot \psi(E)$$

$$\Rightarrow (E-3)^2 = (E-3)^2 \cdot \psi(E)$$

$$\Rightarrow \psi(E) = 1$$

$$PS = {}^nC_2 3^{n-2}$$

$$a_n = a_n^{(H)} + a_n^{(P)}$$

$$= (C_1 + C_2 n)3^n + {}^nC_2 3^{n-2}$$

$$C_1 = 1, C_2 = -1/3$$

A $a_n = 3^{n-2} [9 + 3n + {}^nC_2]$

B $a_n = 3^{n-2} [9 - 3n - {}^nC_2]$

C $a_n = 3^{n-2} [9 - 3n + {}^nC_2]$

D $a_n = 3^{n-2} [9 + 3n - {}^nC_2]$

[MCQ] $t^2 - 2t + 1 = 0$ $a_n^{(H)} = c_1 + c_2 n$
 $(t-1)^2 = 0, t = 1, 1$ $c_1 = 1, c_2 = \frac{1}{3}$

$a_n = 1 + \frac{1}{3}n + \left(\frac{n}{6} + \frac{1}{2}\right)n^2$ 

#Q. Solve the recurrence relation $a_n - 2a_{n-1} + a_{n-2} = n$
 subjected to the initial conditions $a_0 = 1, a_1 = 2$

$f(n) = n = 1^n(n) [b=1]$

$\phi(b) = \phi(1) = 0$

$a_n^{(p)} = 1^n (A_n + B)n^2 = \left[(A_n + B)n^2 - 2 \left\{ (A(n-1) + B)(n-1)^2 \right\} + \left\{ (A(n-2) + B)(n-2)^2 \right\} \right] = n$

A $a_n = 1 + \frac{n}{3} + \left(\frac{n}{6} + \frac{1}{2}\right)n^2$

$n=0, -3A+B=0$
 $n=1, 2B=1$

B $a_n = 1 - \frac{n}{3} + \left(\frac{n}{6} + \frac{1}{2}\right)n^2$

$A = \frac{1}{6}, B = \frac{1}{2}$

$a_n^{(p)} = \left(\frac{1}{6}n + \frac{1}{2}\right)n^2$

C $a_n = 1 + \frac{n}{3} - \left(\frac{n}{6} + \frac{1}{2}\right)n^2$

D $a_n = 1 + \frac{n}{3} + \left(\frac{n}{6} - \frac{1}{2}\right)n^2$

[MCQ]

$$t^2 - 3t - 10 = 0 \quad a_n(t) = c_1 \underbrace{(5)^n + c_2 (-2)^n}$$

$$\Rightarrow t = 5, -2$$



#Q. Solve the recurrence relation $a_n - 3a_{n-1} - 10a_{n-2} = (2n+1) \cdot 5^n$ subjected to the initial conditions $a_0 = 1, a_1 = 2$

$$a_n^{(p)} = 5^n (An+B)n$$

A

$$a_n = \left(\frac{156}{133}\right) 5^n + \frac{23}{133} (-2)^n + 5^n \left(\frac{n}{38} + \frac{5}{38}\right)n$$

$$5^n (An+B)n - 3 \cdot 5^{n-1} \{A(n-1)+B\}$$

B

$$a_n = \left(\frac{156}{133}\right) 5^n - \frac{23}{133} (-2)^n - 5^n \left(\frac{n}{38} + \frac{5}{38}\right)n$$

$$-10 \left[(A(n-2)+B) n-2 \right]^{(n-1)}$$

C

$$a_n = \left(\frac{156}{133}\right) 5^n - \frac{23}{133} (-2)^n + 5^n \left(\frac{n}{38} + \frac{5}{38}\right)n$$

$$n=0, -5A+B=0$$

$$n=1, 3A+7B=1$$

$$A = \frac{1}{38}, B = \frac{5}{38}$$

D

$$a_n = \left(\frac{156}{133}\right) 5^n - \frac{23}{133} (2)^n + 5^n \left(\frac{n}{38} + \frac{5}{38}\right)n$$

$$a_n = \left(\frac{156}{133}\right) 5^n - \frac{23}{133} (-2)^n + 5^n \left(\frac{n}{38} + \frac{5}{38}\right)n$$

[MCQ] $t^2 - 11t + 24 = 0$

$t = 3, 8$

$$a_n(t) = C_1(3)^n + C_2(8)^n$$

$$\Rightarrow C_1 = \frac{77}{73}, C_2 = -\frac{4}{73}$$



#Q. Solve the recurrence relation $a_n = 11a_{n-1} - 24a_{n-2} + 3^n$ subjected to the initial conditions $a_0 = 1, a_1 = 2$

$$a_n = \left(\frac{77}{73}\right)3^n - \frac{4}{73}(8)^n - \frac{n}{5}3^{n-1}$$

A

$$a_n = \left(\frac{77}{73}\right)3^n - \frac{4}{73}(8)^n - \frac{n}{5}(3)^{n-1}$$

$$F(n) = 3^{n+2} = 9 \cdot 3^n \rightarrow b$$

$$a_n(p) = 9. \text{ Sol of } \left(\frac{3^n}{\phi(E)}\right)$$

B

$$a_n = \left(\frac{77}{73}\right)3^n + \frac{4}{73}(8)^n - \frac{n}{5}(3)^{n-1}$$

$$\phi(b) = \phi(3) = 0$$

$$\phi(E) = (E-3) \psi(E)$$

C

$$a_n = \left(\frac{77}{73}\right)3^n - \frac{4}{73}(8)^n + \frac{n}{5}(3)^{n-1}$$

$$(E-3)(E-8) = (E-3)\psi(E)$$

$$\psi(E) = E-8$$

D

$$a_n = \left(\frac{77}{73}\right)3^n - \frac{4}{73}(8)^n - \frac{2n}{5}(3)^{n-1}$$

$$\psi(3) = 3-8 = -5$$

$$a_n(p) = -\frac{1}{5} n C_1 3^{n-1}$$



THANK - YOU